

MARBLEX: Building a smooth data flow between blockchain, the cloud, and gaming

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ABOUT MARBLEX

Netmarble built a system using Google Cloud to directly connect MARBLEX's data, processed through blockchain mainframe, to the game ecosystem.



About MarbleX

Netmarble has been leading the Korean gaming market since its establishment in 2000. It is leading innovation within the industry through the release of numerous blockbuster titles such as 'Everyone's Marble,' 'Seven Knights,' 'Marvel Future Fight,' and 'Lineage 2 Revolution.' As the market center shifted to mobile, it expanded its scope to the global market, establishing itself as a global gaming company, with more than 70% of its revenue coming from the overseas market.

Industries: Games

Google Cloud results

- Eases development environment with a clear and familiar interface
- Provides convenience for infrastructure design, advancement, and management
- Completes an open blockchain ecosystem strategy with seamless data interoperability

Build seamless pipeline between blockchain data game and open ecosystem

Location: South Korea

Load Balancing

Google Cloud (<https://cloud.google.com/>)

Compute Engine

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Armor

Cloud Load Balancing

(<https://cloud.google.com/load-balancing/>)

Compute Engine

(<https://cloud.google.com/compute/>)

Google Cloud Armor

(<https://cloud.google.com/armor/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Since its establishment in 2000, Netmarble has led the growth of the Korean game market with innovative games. It has produced global box office hits enjoyed by gamers around the world, such as 'Everyone's Marble,' 'Seven Knights,' 'Marvel Future Fight,' and 'Lineage 2 Revolution.' With the dawn of the mobile game era, Netmarble's stage has expanded to the world, and more than 70% of its

total sales are generated overseas, expanding its global market.

In line with the growing demand for in-game blockchains, Netmarble is exploring new ways to incorporate blockchain, such as owning and sharing valuable in-game assets. As such, the company created **MARBLEX** (<https://www.marblex.io/en>), to serve as the ecosystem that manages all of its blockchains.

Building a blockchain-based gaming ecosystem

"MARBLEX is a blockchain-based gaming ecosystem built and operated by Netmarble. It was developed based on Netmarble's own key currency, 'MBX,' and serves as a foundation for Netmarble's games to safely and freely integrate with blockchain technology," says Head of MARBLEX Platform Development Office, Yoo Seok-Min.

Yoo says blockchain is linked to games in many ways, and Netmarble wanted to create a blockchain ecosystem that grows together with development studios freely utilizing blockchain ideas that can make games more enjoyable.

The first idea to be served was in NFTs. Through MARBLEX, in-game items can be directly owned by users with NFT-based tokens, and transactions of items between users can be made more accurately and safely. This increases the reliability of the service by being freed from problems that have

plagued gamers for a long time, such as item duplication or loss.

Blockchain API that connects blockchain data easily and reliably

Blockchain is very important because processing speed and reliability are essential requirements. Netmarble built MARBLEX based on Kakao's Klaytn blockchain mainnet and connected it to Google Cloud (<https://cloud.google.com/>) in a hybrid form. When basic data processing, such as storage and movement, is performed on the blockchain, the process of actually connecting the result to the game ecosystem is a structure that is performed through Google Cloud.

The blockchain part consists of three layers: game service chain, bridge chain, and public chain. Each chain optimizes data transmission so that the processing speed is not affected even when users are crowded, and the data flow between the MARBLEX token internal ecosystem and the external ecosystem is processed smoothly. As this multi-chain is processed within the Klaytn mainnet, it looks like a black box that only exchanges data from the outside.

The processing result is met with the game world through the Blockchain API and Batch API of the Google Cloud. The Blockchain API handles the interactions and seamless transactions between your application and the blockchain. The NFT data safely processed in this way is transferred to the Google Cloud. It is seamlessly transferred and

reflected in the game, and the existing game and item experience, such as utilization and trading, can be continued.

"Google Cloud provides a data flow that seamlessly imports and processes data from the blockchain. In particular, the Blockchain API provides a familiar interface, so game developers who are somewhat new to blockchain can immediately use it for development," says Yoo.

Data imported from the blockchain is immediately protected by Google Cloud Armor (<https://cloud.google.com/armor>), which blocks external attacks, and is moved to a virtual machine in Compute Engine (<https://cloud.google.com/compute>), where it is processed in real time. BigQuery (<https://cloud.google.com/bigquery>) and Cloud Load Balancing (<https://cloud.google.com/load-balancing>) smoothly contain the data flowing in, and send and receive the data required by the game stably. Netmarble also plans to introduce automated Pub/Sub (<https://cloud.google.com/pubsub>) to further advance the system. This will allow users to build automated data pipelines between the blockchain and Google Cloud.

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Development Office,
Netmarble

**A data processing environment
that connects 'openness' and
'ecosystem'**

The core of MARBLEX is to have a blockchain-based ecosystem. To that end, MARBLEX has been designed as an open ecosystem. In the future, Netmarble plans to convert the cloud environment currently used by its partners for development, to Google Cloud, to create a more stable environment while ensuring consistency and convenience in management. A public chain has also been created on the mainnet and a link for external connection has been created. Netmarble also plans to transfer the cloud environment utilized by external MARBLEX collaborators to Google Cloud, to increase cloud utilization and migration efficiency.

Data processed on the blockchain could be quickly stored in BigQuery. API services that help send and receive this data will be upgraded to a Google Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine>) (GKE) based environment in the future. Through this, the team plans to secure system stability and more flexible agility in response to unexpected changes in the external environment.

"The infrastructure composition of Google Cloud is simple and clear by default. The structures that were initially conceived in the design were made into infrastructure as they were, and the architecture was completed as easily imagined. The basic framework was established very quickly, and the system was advanced as the necessary parts were added as managed services," says Yoo.

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Netmarble Director Suk Seok-Min emphasized that managed services can be built in a short period of time because managed services reflect various scenarios, and play an appropriate role in the required environment rather than simply implementing functions.

Since Netmarble plans to operate MARBLEX as a platform that supports publishing games released by Netmarble, scalability and interoperability of data are also essential. BigQuery, which acts as a data warehouse, seamlessly connects to all data flows, and easily connects to the desired service to quickly contain data and perform necessary processing immediately. Suk explains that during the development and operation process, he felt that all data flowed safely, quickly, and smoothly within Google Cloud.

"Google Cloud serves as the core infrastructure of MARBLEX. Beyond simple linkage, we plan to implement application-level services such as wallets, order books, and NFTs on Google Cloud. Google Cloud has served as a strong partner for MARBLEX's basic strategy of 'publishing,' connecting blockchains quickly and safely, and connecting various games with blockchains," concludes Suk.

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